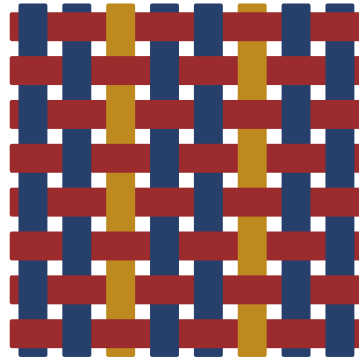




Information Theory

Lecture 2 · from codewords to typical sequences

Oxford B8.4 · 2024--25

fill-in notes from handwritten notes

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dyed in indigo, madder & weld; ruled in iron-gall. woven on LOOM.

THE ONE IDEA, AND THE MAP

□□□□

A source code replaces each source symbol by a finite string. Its quality has two sides: **efficiency**, measured by expected length, and **decodability**, tested after codewords are concatenated. Kraft-McMillan controls which lengths fit; the AEP then shows why entropy is the natural scale for coding long i.i.d. blocks.

The map so far

stop	object	question
1	$C : \mathcal{X} \rightarrow \mathcal{D}^*$	Which finite string represents each source symbol?
2	$L(C) = \mathbb{E}[\ell(X)]$	How many code symbols are used on average?
3	$C^* : \mathcal{X}^* \rightarrow \mathcal{D}^*$	Can concatenated codewords still be decoded uniquely?
4	$\sum_x D^{-\ell(x)} \leq 1$	Which length lists can belong to a decodable code?
5	$-n^{-1} \log p(X^n) \rightarrow H(X)$	Where does the probability mass of a long block concentrate?

Conventions

The source alphabet $\mathcal{X}$ is finite, and $\mathcal{D}^*$ is the set of all finite-length strings over a D -ary code alphabet $\mathcal{D}$. Most examples use $\mathcal{D} = \{0, 1\}$.

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LECTURE 2 · SOURCE CODING

Assign strings to symbols, average their lengths under the source law, and then ask the decisive question: does concatenation preserve enough structure to undo the encoding? For long i.i.d. blocks, identify the typical sequences that carry almost all probability and reveal entropy as the exponential counting rate.

■ 1 CODES, LENGTH, AND DECODABILITY

□□□□ A codeword names one symbol; an extended code must preserve a whole sequence.

Coding begins with a map from source symbols to finite strings. Short strings save space, but distinct symbols and distinct source sequences must remain distinguishable. This is why expected length and decodability have to be studied together.

From a source symbol to a codeword

■ **Definition 1.1** (source code). Let X be a random variable with finite alphabet $\mathcal{X}$. A D -ary source code is a map

$$C : \mathcal{X} \longrightarrow \mathcal{D}^*,$$

where $\mathcal{D}^*$ is the set of finite-length strings over a D -ary alphabet $\mathcal{D}$. The string $C(x)$ is the **codeword** of x , and $\ell(x) = |C(x)|$ is its length.

For instance, if $\mathcal{X} = \{\text{red}, \text{blue}\}$ and $\mathcal{D} = \{0, 1\}$, one may set $C(\text{red}) = 00$ and $C(\text{blue}) = 11$.

■ **Definition 1.2** (expected length). If X has probability mass function p , the expected length of C is

$$L(C) = \mathbb{E}[\ell(X)] = \sum_{x \in \mathcal{X}} p(x) \ell(x).$$

■ **Example 1.3** (an entropy-matched binary code). Let $\mathcal{X} = \{1, 2, 3, 4\}$, with probabilities and codewords

x	1	2	3	4
$p(x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$
$C(x)$	0	10	110	111

The codeword lengths are 1, 2, 3, 3, so

$$L(C) = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = \underline{\frac{7}{4}} \text{ bits.}$$

Since $\ell(x) = -\log_2 p(x)$ for every symbol here,

$$H(X) = \sum_x p(x) \log_2 \frac{1}{p(x)} = \underline{\frac{7}{4}} \text{ bits} = L(C).$$

Your turn. Why does the equality $L(C) = H(X)$ require no averaging trick in this example? The equality holds term by term: each length satisfies $\ell(x) = -\log_2 p(x)$. Multiplying by $p(x)$ and summing therefore gives $L(C) = H(X)$.

Three levels of decodability

Definition 1.4 (*nonsingular code*). A code C is **nonsingular** if different source symbols have different codewords:

$$x \neq x' \implies C(x) \neq C(x').$$

Equivalently, C is injective.

Nonsingularity only protects one symbol at a time. To encode a sequence, extend C by concatenation:

$$C^* : \mathcal{X}^* \longrightarrow \mathcal{D}^*, \quad C^*(x_1 x_2 \cdots x_n) = C(x_1) C(x_2) \cdots C(x_n).$$

Definition 1.5 (*uniquely decodable code*). A code is **uniquely decodable** if its extension C^* is nonsingular. Thus every encoded string has at most one source sequence that produced it.

Definition 1.6 (*prefix code*). A code is a **prefix code**, or instantaneous code, if no codeword is a prefix of any other codeword.

Every prefix code is uniquely decodable, and every uniquely decodable code is nonsingular:

$$\{\text{prefix codes}\} \subset \{\text{uniquely decodable codes}\} \subset \{\text{nonsingular codes}\}.$$

The converses need not hold.

Example 1.7 (*uniquely decodable but not prefix*). For $\mathcal{X} = \{A, B, C, D\}$, consider

$$C(A) = 0, \quad C(B) = 01, \quad C(C) = 011, \quad C(D) = 111.$$

This is not a prefix code because 0 is a prefix of both 01 and 011. Nevertheless, it is uniquely decodable. Removing an initial 0 from a longer codeword leaves only 1 or 11; comparing these suffixes with 111 swaps $1 \leftrightarrow 11$ and never produces the empty suffix. Hence no two source sequences have the same concatenation.

Your turn. The code $\{0, 01, 10\}$ is nonsingular but not uniquely decodable. Exhibit two different source sequences with the same encoded string. *AC and BA both encode as 010.*

■ 2 KRAFT--MCMILLAN: WHEN LENGTHS FIT

□□□□ A decodable code spends a finite amount of room in the code tree.

The Kraft sum turns a list of codeword lengths into a capacity test. Unique decodability forces the sum to be at most one; conversely, every integer length list that passes this test can be packed into a prefix code.

Trigger. Use this test before trying to draw codewords: impossible length lists fail immediately.

The two directions

direction	meaning
necessity	uniquely decodable $\implies \sum_x D^{-\ell(x)} \leq 1$
sufficiency	integer lengths with $\sum_x D^{-\ell(x)} \leq 1 \implies$ a prefix code with exactly those lengths

Theorem 2.1 (*Kraft–McMillan inequality*). Let $C : \mathcal{X} \rightarrow \mathcal{D}^*$ be a D -ary uniquely decodable code, and write $\ell_x = |C(x)|$. Then

$$\sum_{x \in \mathcal{X}} D^{-\ell_x} \leq 1.$$

Conversely, if positive integers $(\ell_x)_{x \in \mathcal{X}}$ satisfy this inequality, then there exists a D -ary prefix code C such that $|C(x)| = \ell_x$ for every $x \in \mathcal{X}$.

Necessity: amplify by concatenating n symbols

Necessary direction. Set

$$K = \sum_{x \in \mathcal{X}} D^{-\ell_x}, \quad \ell_{\min} = \min_x \ell_x, \quad \ell_{\max} = \max_x \ell_x.$$

For a source string $x^n = (x_1, \dots, x_n)$, its encoded length is $\ell(x^n) = \sum_{i=1}^n \ell_{x_i}$. Expanding the n th power gives

$$K^n = \sum_{x^n \in \mathcal{X}^n} D^{-\ell(x^n)} = \sum_{k=n\ell_{\min}}^{n\ell_{\max}} A_k D^{-k},$$

where A_k is the number of source strings in $\mathcal{X}^n$ whose encoded output has length k .

Unique decodability makes these A_k output strings distinct. There are only D^k strings of length k over $\mathcal{D}$, so $A_k \leq D^k$. Therefore

$$K^n \leq \sum_{k=n\ell_{\min}}^{n\ell_{\max}} 1 = n(\ell_{\max} - \ell_{\min}) + 1.$$

Taking n th roots and then letting $n \rightarrow \infty$ yields $K \leq 1$. ■

Your turn. Where exactly did unique decodability enter the proof? It was used to turn source-string counting into output-string counting: distinct source strings must give distinct encoded strings, so among the D^k possible strings of length k we can have at most $A_k \leq D^k$.

Sufficiency: pack adjacent D -ary intervals

Sufficient direction. Relabel $\mathcal{X} = \{1, \dots, M\}$ so that $\ell_1 \leq \ell_2 \leq \dots \leq \ell_M$, and define

$$r_m = \sum_{i=1}^{m-1} D^{-\ell_i}, \quad 1 \leq m \leq M.$$

The Kraft inequality gives $0 \leq r_m < 1$. Since $\ell_i \leq \ell_m$ for $i < m$, $D^{\ell_m} r_m$ is an integer, so r_m has a terminating base- D expansion with at most ℓ_m digits. Let $C(m)$ be its first ℓ_m digits, padding with zeros when necessary.

To see that the resulting code is prefix-free, take $m < n$. By construction,

$$r_n - r_m = \sum_{i=m}^{n-1} D^{-\ell_i} \geq D^{-\ell_m}.$$

But every base- D fraction whose first ℓ_m digits equal $C(m)$ lies in the half-open interval $[r_m, r_m + D^{-\ell_m})$. Thus r_n cannot begin with $C(m)$, so $C(m)$ is not a prefix of $C(n)$. Hence C is a prefix code with the prescribed lengths. ■

Example 2.2 (a tight binary packing). For lengths $(1, 2, 3, 3)$,

$$2^{-1} + 2^{-2} + 2^{-3} + 2^{-3} = 1.$$

The construction uses left endpoints $0, \frac{1}{2}, \frac{3}{4}, \frac{7}{8}$, whose binary expansions give the prefix code 0, 10, 110, 111.

Your turn. Could a binary uniquely decodable code have lengths $(1, 2, 2)$? Compute the Kraft sum and state the verdict.

$$2^{-1} + 2^{-2} + 2^{-2} = \underline{1}.$$

Yes. The Kraft sum is 1, so the converse constructs a prefix code; one choice is 0, 10, 11.

? Why does the n th-root limit force the Kraft sum to be at most one?

3 AEP AND TYPICAL SEQUENCES

□□□□ A long random block behaves as though its likely outcomes were nearly equiprobable.

The asymptotic equipartition property turns entropy into a geometric picture: almost all probability lies on roughly $2^{nH(X)}$ length- n sequences, and each of those sequences has probability roughly $2^{-nH(X)}$.

Trigger. Use typical sets when a one-symbol distribution must be converted into a statement about long blocks.

A Bernoulli preview

Example 3.1 (*counting zeros and ones*). Let $X \sim \text{Bernoulli}(p)$ and let $X_1, \dots, X_n$ be i.i.d. copies. If a realization $x^n = (x_1, \dots, x_n)$ contains $N_1(x^n)$ ones, then

$$p_{X^n}(x^n) = p^{N_1(x^n)}(1-p)^{n-N_1(x^n)}.$$

The law of large numbers says $N_1(X^n)/n \rightarrow p$ in probability. Thus, for most long realizations,

$$\begin{aligned} -\frac{1}{n} \log p_{X^n}(X^n) &\approx -p \log p - (1-p) \log(1-p) \\ &= \underline{H(X)}. \end{aligned}$$

Equivalently, a typical realization has probability about $2^{-nH(X)}$.

Three modes of random-variable convergence

Definition 3.2 (*convergence in probability*). Let $Y_1, Y_2, \dots$ and Y be real-valued random variables on the same probability space. We say that Y_n **converges in probability** to Y , and write $Y_n \xrightarrow{P} Y$, if for every $\epsilon > 0$,

$$\mathbb{P}(|Y_n - Y| > \epsilon) \rightarrow \underline{0} \quad \text{as } n \rightarrow \infty.$$

Thus convergence in probability controls the probability of a fixed-size error; it does not assert that every sample path eventually stays close.

Definition 3.3 (*mean-square convergence*). Suppose $Y_1, Y_2, \dots$ and Y are square-integrable. We say that Y_n **converges in mean square**, or in L^2 , to Y if

$$\mathbb{E}[|Y_n - Y|^2] \rightarrow 0.$$

The notation is $Y_n \xrightarrow{L^2} Y$. This asks the average squared error, not merely the probability of a large error, to vanish.

Definition 3.4 (*almost-sure convergence*). We say that Y_n **converges almost surely**, or with probability one, to Y if

$$\mathbb{P}\left(\left\{\omega : \lim_{n \rightarrow \infty} Y_n(\omega) = Y(\omega)\right\}\right) = 1.$$

The notation is $Y_n \xrightarrow{\text{a.s.}} Y$. Thus, outside a single probability-zero exceptional set, the numerical sequence $Y_n(\omega)$ converges to $Y(\omega)$ along every remaining sample path.

The implications needed most often are

$$Y_n \xrightarrow{L^2} Y \implies Y_n \xrightarrow{P} Y, \quad Y_n \xrightarrow{\text{a.s.}} Y \implies Y_n \xrightarrow{P} Y.$$

The first follows from $\mathbb{P}(|Y_n - Y| > \epsilon) \leq \mathbb{E}[|Y_n - Y|^2]/\epsilon^2$. In general, mean-square and almost-sure convergence do not imply one another without extra assumptions.

In the AEP, $Y_n = -n^{-1} \log p_{X^n}(X_1, \dots, X_n)$ and the limit $Y = H(X)$ is constant; the theorem asserts convergence in probability.

Theorem 3.5 (*weak asymptotic equipartition property*). Let X be a discrete random variable on a finite alphabet, and let $X_1, X_2, \dots$ be i.i.d. copies of X . Then

$$-\frac{1}{n} \log p_{X^n}(X_1, \dots, X_n) \longrightarrow H(X) \quad \text{in probability.}$$

Proof. On the support of X , define $Z_i = -\log p_X(X_i)$. Independence gives

$$-\log p_{X^n}(X_1, \dots, X_n) = \sum_{i=1}^n -\log p_X(X_i) = \sum_{i=1}^n Z_i.$$

The variables Z_i are i.i.d. and

$$\mathbb{E}[Z_i] = \sum_x p_X(x) \log \frac{1}{p_X(x)} = H(X).$$

The weak law of large numbers applied to $n^{-1} \sum_i Z_i$ proves the claim. ■

Your turn. What is the one-symbol random variable to which the law of large numbers is applied in the AEP proof? **Apply the law of large numbers to the surprise $Z = -\log p_X(X)$. Its expectation is exactly $H(X)$.**

Weak typicality: make "about" precise

Definition 3.6 (*weakly typical set*). For $\epsilon > 0$, the length- n weakly typical set is

$$T_{n,\epsilon} = \left\{x^n \in \mathcal{X}^n : \left| -\frac{1}{n} \log p_{X^n}(x^n) - H(X) \right| \leq \epsilon \right\}.$$

Its members are the ϵ -typical sequences of length n .

Theorem 3.7 (*typical-set properties*). For every $\epsilon > 0$, all sufficiently large n satisfy:

1. every $x^n \in T_{n,\epsilon}$ obeys

$$2^{-n(H(X)+\epsilon)} \leq p_{X^n}(x^n) \leq 2^{-n(H(X)-\epsilon)};$$

2. $\mathbb{P}(X^n \in T_{n,\epsilon}) > 1 - \epsilon$;

3. the cardinality is bounded by

$$(1 - \epsilon)2^{n(H(X)-\epsilon)} < |T_{n,\epsilon}| \leq 2^{n(H(X)+\epsilon)}.$$

The first property follows from the definition for every n ; only the latter two require n to be large.

Proof. The first claim is obtained by exponentiating the defining inequality. The AEP gives

$$\mathbb{P}(X^n \notin T_{n,\epsilon}) = \mathbb{P}\left(\left| -\frac{1}{n} \log p_{X^n}(X^n) - H(X) \right| > \epsilon\right) \rightarrow 0,$$

which proves the second claim.

For the upper cardinality bound, use the lower probability bound on each typical sequence:

$$1 \geq \sum_{x^n \in T_{n,\epsilon}} p_{X^n}(x^n) \geq |T_{n,\epsilon}| 2^{-n(H(X)+\epsilon)}.$$

For the lower cardinality bound, use the upper probability bound together with the mass estimate:

$$1 - \epsilon < \sum_{x^n \in T_{n,\epsilon}} p_{X^n}(x^n) \leq |T_{n,\epsilon}| 2^{-n(H(X)-\epsilon)}.$$

Rearranging gives the stated bounds. ■

Your turn. Complete the AEP slogan by combining the probability and cardinality bounds: there are about $\underline{2^{nH(X)}}$ typical sequences, each with probability about $\underline{2^{-nH(X)}}$.

The smallest set carrying almost all mass

Definition 3.8 (*minimum high-probability set*). For $0 < \delta < 1$, let $S_m^\delta \subseteq \mathcal{X}^m$ be a set of minimum cardinality subject to

$$\mathbb{P}(X^m \in S_m^\delta) \geq 1 - \delta.$$

This is the “strong” viewpoint in the handwritten notes: keep only the most probable sequences needed to capture the prescribed mass.

Proposition 3.9 (*exponential equivalence of typical sets*). Let (δ_j) be any positive sequence with $\delta_j \rightarrow 0$. Then

$$\lim_{j \rightarrow \infty} \limsup_{m \rightarrow \infty} \left| \frac{1}{m} \log \frac{|S_m^{\delta_j}|}{|T_{m, \delta_j}|} \right| = 0.$$

Thus the minimum high-probability set and the weakly typical set may differ in their individual members, but not in their exponential size as the tolerance shrinks.

Proof. For each fixed $0 < \delta < 1$, the AEP and the typical-set bounds imply

$$\lim_{m \rightarrow \infty} \frac{1}{m} \log |S_m^\delta| = H(X), \quad H(X) - \delta \leq \liminf_m \frac{1}{m} \log |T_{m, \delta}| \leq \limsup_m \frac{1}{m} \log |T_{m, \delta}| \leq H(X) + \delta.$$

For the first limit, an upper bound comes from choosing a typical set; a lower bound follows by intersecting any set of mass at least $1 - \delta$ with an η -typical set and then letting $\eta \downarrow 0$. Hence the normalized logarithm of the ratio has absolute limsup at most δ . Substituting $\delta = \delta_j$ and letting $j \rightarrow \infty$ proves the result. ■

? How do "probability per typical sequence" and "number of typical sequences" multiply back to one?